

Elements of Position

Companion: Verification and Models

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A note. This is not a numbered part of the series. It is documentation: a description of the code that verifies claims made in Papers I–III at a depth their own printed tables cannot show, and that gives concrete visual form to the plan/section/elevation reading Paper IV names but does not itself draw. Nothing here is a philosophical argument, and nothing here extends the series’ claims. Where this companion’s tooling turned up structure unrelated to anything the papers argue, that material is deliberately not included here; it is held in a separate, clearly labeled exploratory record.

Why this exists

Papers I through III make claims that hold at every finished n : the geometric mean locks at 1, the marriage of Process A and Process B is exact only at $n = 5$, the Fibonacci docking error shrinks as the count runs. Each paper demonstrates this with a printed table running to $n = 16$ — a table can’t run further and still be read. That is a real limitation, not a rhetorical one: “one run is every run” (Paper III) is a claim about all n , and sixteen rows support it by example, not by exhaustion.

This companion exists to close that gap: two scripts that carry every claim out to a depth no page could hold, self-test the results on every execution, and give a live, orbitable model of the two-process geometry Paper III describes only in prose and static figures.

What is included, and what is deliberately not

Two scripts, described in §3–§4. Both are built on formulas ported verbatim from `Elements_of_Position-Drawings.py`, the script every paper already cites — this companion is a second reading of the same geometry, not a second geometry.

During the construction of these tools, several structures were found that are true, checkable, and unrelated to anything the papers claim — among them a connection between the vertex structure explored in Model 2 and Euler’s totient function, and a further connection to cyclotomic polynomials. Those findings are real and documented, but held in a separate exploratory record: they support nothing written in Papers I–V. Keeping that boundary explicit is itself part of the discipline the papers ask of themselves.

Verification.py — Papers I, II, III

Ports `row_for_n`, `count_process`, and `phi_process` directly from the drawings script, then runs the master sequence to $n = 10,000$ by default (configurable), producing:

- `measurements.csv` — one row per n : product, geometric mean, arithmetic mean, cross-ratio $-n$, half-walk $n\pi/2$, chord, apothem, and `marriage_gap` $= 2\cos(\pi/n) - \varphi$ (Paper III’s closing lemma, generalized across the full range).
- `docking_events.csv` — the Fibonacci docking table from Paper III, Proposition 1, computed two ways per row: the paper’s own formula $(\frac{1}{2}|F_k\Phi - F_{k-1}|)$ and a closed form derived here via Binet’s identity, $\Phi^{k+2}/2$, confirmed to agree to 44+ decimal digits under arbitrary-precision arithmetic. The paper’s formula subtracts two large, nearly equal numbers — ordinary floating-point cancellation,

visible directly in the `cancellation_error` column, growing from $\sim 10^{-17}$ at $k = 1$ to $\sim 10^{-13}$ by $k = 18$. The closed form has no such term and is used as the reported value.

- `live_model.html` — Process A and Process B in one orbitable 3D scene, with a configurable arrival point for B, Fibonacci dockings marked, and the $n = 5$ marriage called out directly.

k	F_k	docking error
1	2	0.118034
5	13	0.017221
10	144	0.001553
15	1597	0.000140
18	6765	0.000033

Table 1: A sample of the docking table at $n = 10,000$ depth. Full table in `docking_events.csv`.

Self-test on every run: `docking_error == Phi**(k+2)/2` across all rows in range, to within 10^{-12} .

Local Circle Models.py — Papers I, III, IV

Three models, matching Paper IV’s own division of one object into plan, section, and elevation:

Model 1 — local circle. One n -gon at a time, built raw from the master sequence: radius $n/2$, chord $n \sin(\pi/n)$, apothem $(n/2) \cos(\pi/n)$, central angle $2\pi/n$, and the isosceles spoke triangle’s base angles. This is Paper III’s worked $n = 5$ example (radius 2.5, central angle 72° , base angles 54° — reproduced exactly by this script) generalized to any finished n .

Model 2 — plan overlay. Every local circle from Model 1, overlaid concentrically at one center — the flat nesting Paper III describes when it says scale does not matter.

Model 3 — elevation stack. The same n -gons, stacked by height $n/2$ instead of flattened to one plane. This height is not an arbitrary choice: Paper I’s own cone construction already lifts the own-origin to height $n/2$, forming the 45–45–90 triangle with vertices $(0,0)$, $(n,0)$, $(n/2, n/2)$. The vertex line running straight down the elevation stack at $\theta = 0$ is that triangle’s hypotenuse, already proved in Paper I. The section cut at the mid-edge angle traces the apothem, $(n/2) \cos(\pi/n)$, which converges onto that same line as n grows — a picture of the Archimedes inequality $n \sin(\pi/n) < \pi < n \tan(\pi/n)$ that Paper I cites in its literature review but does not itself draw.

Self-tests on every run: the relative gap $1 - (\text{apothem}/\text{radius})$ matches $\pi^2/(2n^2)$ to within 1% for $n \geq 500$; the inscribed and circumscribed Archimedes bounds hold on every side, for every n , with no exceptions.

Reproducing this companion

```
python "Elements of Position - Verification.py" --nmax 10000
python "Elements of Position - Local Circle Models.py" --nmax 1000
```

Both scripts print a `self-test:` block on completion. A run that does not print all `PASS` lines should not be trusted, and neither should any figure produced by it.

Scope

This companion verifies. It does not extend. Nothing here should be read as a sixth part of the series, and nothing here bears on Paper IV’s analytic material (the derived-unit, vacuum/trace, and midline discussion) — that content is explicitly a naming choice within the paper itself, not a numerical claim, and no tool in this companion touches it either way.